

1. Connectivness of Graphs

Def. 1. Vertices $u, v \in V, u \neq v$ are called **connected** if there is a $u - v$ -path in G .

Def. 2. A graph $G = (V, E)$ is **connected** if $\forall u, v \in V, u \neq v$ a $u - v$ -path in G exists.

Def. 3. Vertices $u, v \in V, u \neq v$ are called **k -connected** if $|V| \geq k + 1$ and

$$\forall X \subseteq V \setminus \{u, v\}$$

$$\text{with } |X| < k,$$

$$u \text{ and } v \text{ are connected in } G[V \setminus X]$$

Def. 4. A graph $G = (V, E)$ is called **k -connected** if $|V| > k$ and

$$\forall X \subseteq V \text{ with } |X| < k$$

$$G[V \setminus X] \text{ is connected}$$

Def. 5. $X \subseteq V \setminus \{u, v\}$ is called a **$u - v$ -separator** if u and v are in different connected components of $G[V \setminus X]$.

Rem. 6. If G contains a vertex v with $\deg(v) < k$, then G cannot be k -connected.

Thm. 7. Menger's Theorem (Vertex Version): The minimum size of a $u - v$ -separator is equal to the maximum number of internally vertex-disjoint $u - v$ -paths. A graph is k -connected iff $\forall u \neq v$ there are k internally vertex-disjoint $u - v$ -paths.

Def. 8. A graph $G = (V, E)$ is called **k -edge-connected** if

$$\forall X \subseteq E \text{ with } |X| < k$$

$$(V, E \setminus X) \text{ is connected}$$

Thm. 9. Menger's Theorem (Edge Version): A graph G is k -edge-connected iff. $\forall u, v \in V, u \neq v$ there are k edge-disjoint $u - v$ -paths.

Thm. 10. It always holds that:

$$\begin{aligned} \text{Vertex Connectivity} &\leq \text{Edge Connectivity} \\ &\leq \text{Minimum Degree } \delta(G) \end{aligned}$$

Lets look at the special case where $k = 1$:

Def. 11. For a connected graph $G = (V, E)$:

- Vertex $v \in V$ is a **cut vertex** iff. $G[V \setminus \{v\}]$ is disconnected.
- Edge $e \in E$ is a **bridge** iff. $(V, E \setminus \{e\})$ is disconnected.

Lem. 12. If $e = \{x, y\} \in E$ is a bridge, then $\deg(x) = 1$ or x is a cut vertex.

Def. 13. Blocks: Define an equivalence relation $\sim$ on E :

$$e \sim f \iff e = f \vee \exists \text{ cycle containing } e \text{ and } f$$

The equivalence classes are called **blocks**.

Note: Two blocks can share at most one vertex, which must be a cut vertex.

Def. 14. Block Graph: Bipartite graph $T = (A \cup B, E_T)$ where: $A = \{\text{cut vertices}\}$ $B = \{\text{blocks}\}$ Edge $\{a, b\} \in E_T \iff a$ is incident to an edge in block b .

Thm. 15. If G is connected, the block graph of G is a tree.

Algorithm 16

```
1 #DFS & Low Values for cut vertices/bridges in O(|V|+|E|)
2 low[v] := #smallest DFS number reachable from v
   by tree edges and <=1 back edge.
3
4 low[v] = min{ dfs[w], min_{(v,w) back} dfs[w],
   min_{(v,w) tree} low[w] }
5
6 #A vertex v is a cut vertex iff:
7 v != root and exists child u with low[u] >= dfs[v]
   ].
8 v = root and has >= 2 children in DFS tree.
9
10 #A tree edge (v,w) is a bridge iff:
11 low[w] >= dfs[v].
```

2. Eulertours

Def. 17. An **Eulertour** is a closed walk in a graph G that contains every edge exactly once.

Thm. 18. A connected graph $G = (V, E)$ contains an Eulertour iff the degree of every vertex is even.

Algorithm 19

```
1 #Find Eulertour in O(|E|) (Runner and Turtle)
2 Runner traverses until hitting a visited vertex (
   cycle found).
3 Turtle follows path until an unvisited neighbor
   is found.
4 Runner is set loose again from there, expanding
   the tour by cycles.
```

3. Hamiltonian Cycles

Def. 20. A **Hamiltonian Cycle** is a cycle containing every vertex exactly once.

Thm. 21. For $m, n \geq 2$, an $n \times m$ grid has a Hamiltonian cycle iff $n \cdot m$ is even.

Thm. 22. A bipartite graph $G = (A \cup B, E)$ with $|A| \neq |B|$ cannot contain a Hamiltonian cycle.

Ex. 23. The n -dimensional hypercube Q_n is bipartite and has a Hamiltonian cycle iff $n \geq 2$.

Thm. 24. Dirac's Theorem: Every graph G with $|V| \geq 3$ and minimum degree $\delta(G) \geq |V|/2$ contains a Hamiltonian cycle.

Algorithm 25

```
1 #DP for Hamiltonian Cycle O(n^2 * 2^n)
2 #P[S, x] = 1 if exists 1-x path containing
   exactly vertices in S.
3 P[S, x] = max { P[S \setminusminus \{x\}, y] | y \in N
   (x) \cap S, y \neq x }
```

3.1 Travelling Salesman Problem (TSP)

The TSP finds a Hamiltonian cycle in K_n with minimal weight (shortest roundtrip).

Thm. 26. The TSP is NP-Complete (reducible from Hamiltonian Cycle).

3.1.1 Metric TSP

For graphs where triangle inequality holds:

$$w(u, v) \leq w(u, w) + w(w, v)$$

There is a 2-approximation algorithm.

Algorithm 27

```
1 \textbf{2-Approximation for Metric TSP}: 1.
   Determine MST T. 2. Double edges of T to
   get Eulerian multigraph T'. 3. Traverse
   Eulertour, skip already visited vertices.
```

Thm. 28. Christofides 1.5-Approximation for Metric TSP: 1. Determine MST T. 2. Find minimal matching M for the odd-degree vertices of T. 3. Determine Eulertour in T with edges of M added.

4. Matchings

Def. 29. An edge set $M \subseteq E$ is a **matching** in $G = (V, E)$ if no vertex is incident to more than one edge from M .

- **Covered:** Vertex v is incident to an edge in M .
- **Perfect:** Every vertex is covered ($|M| = |V|/2$).
- **Maximal:** Cannot be extended by adding edges.
- **Maximum:** No matching with more edges exists.

Algorithm 30

```
1 #Greedy Algorithm (Leads to Maximal Matching)
2 #Size of M is at least half the size of a maximum
   matching.
3 M = empty set
4 While E is not empty:
5   Choose arbitrary edge e from E, add to M
6   Remove all edges incident to e's endpoints from
   E
```

Thm. 31. Hall's Marriage Theorem: A bipartite graph $G = (A \cup B, E)$ contains a matching of cardinality $|M| = |A|$ iff:

$$\forall X \subseteq A : |X| \leq |N(X)|$$

Thm. 32. Frobenius Corollary: Every k -regular bipartite graph contains a perfect matching. In fact, it is a union of perfect matchings. A perfect matching can be found in $O(|E|)$ time if the graph is k - or 2^k -regular.

Def. 33. An **M -augmenting path** alternates between edges in M and not in M , starting and ending in uncovered vertices.

Thm. 34. Berge's Theorem (1957): A matching M is maximum iff there is no M -augmenting path.

Algorithm 35

```
1 \textbf{Hopcroft-Karp} ($O(|V|^{1/2}|E|)$ time
   for bipartite graphs): While there is an
   augmenting path, find a maximal set of
   vertex-disjoint shortest augmenting paths
   using BFS, and augment $M$ along all of
   them.
```

5. Coloring

Def. 36. A k -coloring of $G = (V, E)$ is a mapping $c : V \rightarrow [k]$ such that $c(u) \neq c(v)$ for all $\{u, v\} \in E$. A **stable set** is a set of vertices where no two are adjacent. The **chromatic number** $\chi(G)$ is the minimum number of colors. $\chi(G) \leq 2 \iff G$ is bipartite.

Thm. 37. For $k \geq 3$, the problem "Is $\chi(G) \leq k$?" is NP-complete. For general graphs all ϵ -approximations are NP-hard.

Algorithm 38

```
1 \textbf{Greedy Coloring} ($\Delta(G) + 1$ colors):
2   \begin{itemize}
3     \item There exists an optimal vertex order
        yielding exactly $\chi(G)$ colors.
4     \item \textbf{Heuristic}: Delete vertices in
        order of minimum degree. If every
        subgraph has a vertex of degree $\le k$,
        this yields $\le k+1$ colors (e.g.
        trees are $2$-colorable).
5   \end{itemize}
6 \textbf{Block-graphs}: If every block can be $k$-
   colored, the graph can be $k$-colored.
```

Thm. 39. Brooks' Theorem: For a connected graph G that is neither a complete graph K_n nor an odd cycle C_{2n+1} :

$$\chi(G) \leq \Delta(G)$$

Can be found in $O(|E|)$ time by: 1) If $\Delta(G) = 2$, 2-color. 2) If $\exists v, \deg(v) < \Delta(G)$ use greedy heuristic. 3) If $\exists$ cut vertex, color blocks. 4) Else remove non-adj. $x, y \in N(v)$ and color recursively.

6. Intro: Probability

Def. 40. A **discrete probability space** is a set of elementary events $\Omega = \{\omega_1, \dots, \omega_n\}$. Every event ω_i has a probability $P[\omega_i] \in [0, 1]$ such that

$$\sum_{\omega_i \in \Omega} P[\omega_i] = 1$$

Rem. 41. A finite space where every event has equal probability is a **Laplace-Space**. For any event E :

$$P[E] = \frac{|E|}{|\Omega|}$$

Def. 42. An **Event** $E \subseteq \Omega$ has probability $P[E] = \sum_{\omega_i \in E} P[\omega_i]$. The complementary event $\bar{E} = \Omega \setminus E$ has probability $P[\bar{E}] = 1 - P[E]$.

Thm. 43. For two events $A, B \subseteq \Omega$: $P[\emptyset] = 0$, $P[\Omega] = 1$, $P[A] \in [0, 1]$. If $A \subseteq B$ then $P[A] \leq P[B]$.

Thm. 44. Union Bound: For events $A_1, \dots, A_n$:

$$P\left[\bigcup_{i=1}^n A_i\right] \leq \sum_{i=1}^n P[A_i]$$

(Equality iff A_i are pairwise disjoint).

Thm. 45. Inclusion-Exclusion Principle: For $n \geq 2$:

$$P\left[\bigcup_{i=1}^n A_i\right] = \sum_{k=1}^n (-1)^{k+1} \sum_{I \subseteq [n], |I|=k} P\left[\bigcap_{j \in I} A_j\right]$$

6.1 Conditional Probability

Def. 46. For $P[B] > 0$, the **conditional probability** of A given B is:

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]}$$

Rem. 47. Always holds: $P[A \mid \Omega] = P[A]$, $P[A \mid A] = 1$, $P[A \mid \bar{A}] = 0$, $P[A \cap B] = P[A \mid B]P[B]$.

Thm. 48. Multiplication-Theorem: If $P\left[\bigcap_{i=1}^{n-1} A_i\right] > 0$:

$$P\left[\bigcap_{i=1}^n A_i\right] = P[A_1] \cdot P[A_2 \mid A_1] \cdot \dots \cdot P[A_n \mid \bigcap_{i=1}^{n-1} A_i]$$

Thm. 49. Theorem of total probability: For pairwise disjoint A_i and $B \subseteq \bigcup_{i=1}^n A_i$:

$$P[B] = \sum_{i=1}^n P[B \mid A_i]P[A_i]$$

Thm. 50. Bayes' Theorem: For exactly defined disjoint A_i , $B \subseteq \bigcup A_i$, and $P[B] > 0$:

$$P[A_i \mid B] = \frac{P[B \mid A_i]P[A_i]}{\sum_{j=1}^n P[B \mid A_j]P[A_j]}$$

6.2 Independence

Def. 51. Two events A, B are **independent** if:

$$P[A \cap B] = P[A]P[B] \text{ or } P[A \mid B] = P[A]$$

Events $A_1, \dots, A_n$ are independent if for all $I \subseteq \{1, \dots, n\}$:

$$P\left[\bigcap_{i \in I} A_i\right] = \prod_{i \in I} P[A_i]$$

Lem. 52. Let $A_1, \dots, A_n$ be events, $A_i^0 = \bar{A}_i$, $A_i^1 = A_i$. They are independent iff $\forall (s_1, \dots, s_n) \in \{0, 1\}^n$:

$$P\left[\bigcap_{i=1}^n A_i^{s_i}\right] = \prod_{i=1}^n P[A_i^{s_i}]$$

Lem. 53. If A, B, C are independent, then so are $A \cap B$ and C , and $A \cup B$ and C .

7. Random Variables

Def. 54. A **random variable** is a map $X : \Omega \rightarrow \mathbb{R}$. The **probability mass function (PMF)** is $f_X(x) = P(X = x)$. The **distribution function (CDF)** is $F_X(x) = P(X \leq x)$.

7.1 Expectation

Def. 55. The expectation of X is defined as:

$$\mathbb{E}[X] := \sum_{\alpha \in W_x} \alpha P(X = \alpha)$$

Thm. 56. For a random variable X with values in $\mathbb{N}_0$:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k)$$

Def. 57. Indicator-variable X_E for $E \subseteq \Omega$: $X_E(\omega) = 1$ if $\omega \in E$, else 0.

$$\mathbb{E}[X_E] = P(E)$$

Thm. 58. Linearity of Expectation: For random variables $X_1, \dots, X_n$ and $a_1, \dots, a_n, b \in \mathbb{R}$:

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i + b\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i] + b$$

Algorithm 59

```
1 #Maximal Stable Set Algorithm
2 S[v] = 1/0 with probability p
3 For all {u,v} in E:
4   if S[u] == 1 and S[v] == 1: S[v] = 0
5 Expected size: $\mathbb{E}[|\text{MSS}|] \geq n \cdot p$
   - $n \cdot p^2$
```

7.2 Variance

Def. 60. For X with $\mathbb{E}[X] = \mu$, the **variance** is:

$$\text{Var}[X] := \mathbb{E}[(X - \mu)^2]$$

Standard deviation $\sigma := \sqrt{\text{Var}[X]}$.

Thm. 61.

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Lem. 62. For $a, b \in \mathbb{R}$: $\text{Var}[aX + b] = a^2 \text{Var}[X]$

Def. 63. $\mathbb{E}[X^k]$ is the k -th **moment**. $\mathbb{E}[(X - \mathbb{E}[X])^k]$ is the k -th **central moment**.

7.3 Multiple & Combined Variables

Def. 64. RVs $X_1, \dots, X_n$ are **independent** iff $\forall a_i \in W_{X_i}$:

$$P[X_1 = a_1, \dots, X_n = a_n] = \prod_{i=1}^n P[X_i = a_i]$$

Thm. 65. If X, Y are independent, $f_{X+Y}(\alpha) = \sum_{\beta} f_X(\beta) f_Y(\alpha - \beta)$.

Rem. 66. Calculation Rules:

- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- If X, Y independent: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- If X, Y independent: $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$

Note $\text{Var}[XY] \neq \text{Var}[X]\text{Var}[Y]$ in general.

Thm. 67. Total Expectation: For pairwise disjoint events $A_1, \dots, A_n$ covering Ω :

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X \mid A_i] P[A_i]$$

Thm. 68. Wald’s Equation: If N and X are independent RVs and $Z := \sum_{i=1}^N X_i$ (where X_i are independent copies of X):

$$\mathbb{E}[Z] = \mathbb{E}[N] \mathbb{E}[X]$$

8. Distributions

Def. 69. Bernoulli(p): $X \in \{0, 1\}$

$$P[X = 1] = p \quad P[X = 0] = 1 - p$$
$$\mathbb{E}[X] = p \quad Var[X] = p(1 - p)$$

Def. 70. Binomial(n, p): $X \in \{0, \dots, n\}$

$$P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$
$$\mathbb{E}[X] = np \quad Var[X] = np(1 - p)$$

Def. 71. Poisson(λ):

$$P[X = k] = \frac{e^{-\lambda} \lambda^k}{k!}$$
$$\mathbb{E}[X] = \lambda \quad Var[X] = \lambda$$

Def. 72. Geometric(p):

$$P[X = k] = p(1 - p)^{k-1}$$
$$\mathbb{E}[X] = \frac{1}{p} \quad Var[X] = \frac{1 - p}{p^2}$$

(Number of throws until first success).

Def. 73. Negative Binomial of order n :

$$P[X = k] = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$$

(Number of throws until n -th success).

9. Bounding Probability

Thm. 74. Markov’s Inequality: For non-negative RV X and $t \geq 0$:

$$P[X \geq t] \leq \frac{\mathbb{E}[X]}{t} \quad \text{or} \quad P[X \geq t \mathbb{E}[X]] \leq \frac{1}{t}$$

Thm. 75. Chebyshev’s Inequality: For any RV X and $t \geq 0$:

$$P[|X - \mathbb{E}[X]| \geq t] \leq \frac{Var[X]}{t^2}$$

9.1 Chernoff Bounds

Applies to sums of independent Bernoulli RVs $X = \sum X_i$ with $P[X_i = 1] = p_i$.

Thm. 76. For $0 \leq \delta \leq 1$:

$$P[X \geq (1 + \delta) \mathbb{E}[X]] \leq e^{-\frac{1}{3} \delta^2 \mathbb{E}[X]}$$
$$P[X \leq (1 - \delta) \mathbb{E}[X]] \leq e^{-\frac{1}{2} \delta^2 \mathbb{E}[X]}$$

For larger deviations ($t \geq 2e \mathbb{E}[X]$):

$$P[X \geq t] \leq 2^{-t}$$

9.2 Target Shooting (Sampling)

To estimate $|S|/|U|$, sample n elements uniformly. Let $Y = \frac{1}{n} \sum Y_i$ be the fraction of samples in S .

Thm. 77. With given $\delta, \epsilon \geq 0$, if $n \geq \frac{3|U|}{|S|\epsilon^2} \ln \frac{2}{\delta}$, then

$$Y \in \left[(1 - \epsilon) \frac{|S|}{|U|}, (1 + \epsilon) \frac{|S|}{|U|} \right]$$

with probability $\geq 1 - \delta$.